

# IoT-enabled digital product passports for real-time lifecycle data in manufacturing.

Manufacturing × Product traceability and digital product passport × IoT platform · OS005 v2 · refreshed 2026-08-18

| ATTRACTIVENESS                   | RIGHT TO WIN                         | COMPETITION                            | SERVICEABLE MARKET                            | HORIZON                                         | PORTFOLIO               |
|----------------------------------|--------------------------------------|----------------------------------------|-----------------------------------------------|-------------------------------------------------|-------------------------|
| <b>55</b><br>is the world moving | <b>61</b><br>can we play, can we win | <b>HIGH</b><br>0 named in the evidence | <b>€2.9bn</b><br>observed evidence · per year | <b>NEXT</b><br>regulation adopted or propose... | <b>L1</b><br>14 signals |

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

## The opportunity

IoT-enabled digital product passports give manufacturers a real-time, machine-readable record of a product's lifecycle, from raw material to end-of-life. This opportunity exists now because the EU's Ecodesign for Sustainable Products Regulation (ESPR) mandates DPPs for electronics and appliances by 2027, creating an urgent compliance need. Orange Business can assemble a sovereign, secure IoT and data platform to help manufacturers meet this regulation while unlocking circular economy value.

## Market opportunity

| Method                                             | Total (TAM)                      | Serviceable (SAM)                | Obtainable (SOM)                | Confidence |
|----------------------------------------------------|----------------------------------|----------------------------------|---------------------------------|------------|
| Bottom-up: enterprises x adoption x contract value | <b>€4.0bn</b><br>€715m – €21.4bn | <b>€2.9bn</b><br>€513m – €15.3bn | <b>€79.8m</b><br>€14.4m – €429m | observed   |
| Observed: tendered contract value, annualised      | <b>€20.9m</b><br>€20.9m – €56.6m | <b>€20.9m</b><br>€20.9m – €56.6m | <b>€586k</b><br>€586k – €1.6m   | observed   |

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

### Bottom-up: enterprises x adoption x contract value

| Factor                                             | Value                                                              | Basis      | Source                                        | As of                  |
|----------------------------------------------------|--------------------------------------------------------------------|------------|-----------------------------------------------|------------------------|
| Enterprises in Manufacturing (EU27_2020)           | 238,122 enterprises                                                | observed   | Eurostat · sbs_sc_ovw                         | 2024                   |
| Enterprises adopting IoT platform                  | 30.0 % of enterprises (10+ employees)                              | observed   | Eurostat · isoc_eb_iotn2 · E_IOT1             | 2021                   |
| Annual value of one engagement                     | €337k per enterprise per year                                      | observed   | TED (Tenders Electronic Daily) · ted          | 2026-05-19..2026-08-05 |
| Size-weighted buyer base                           | 39,220 enterprise-equivalents at large-enterprise engagement value | assumption | config/sizing.yaml · size_class_value_weights | size-2026-08-a         |
| Assumed obtainable share of the serviceable market | 2.8 % of SAM                                                       | assumption | config/sizing.yaml · obtainable_share         | size-2026-08-a         |

**Read this before quoting the number.** Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

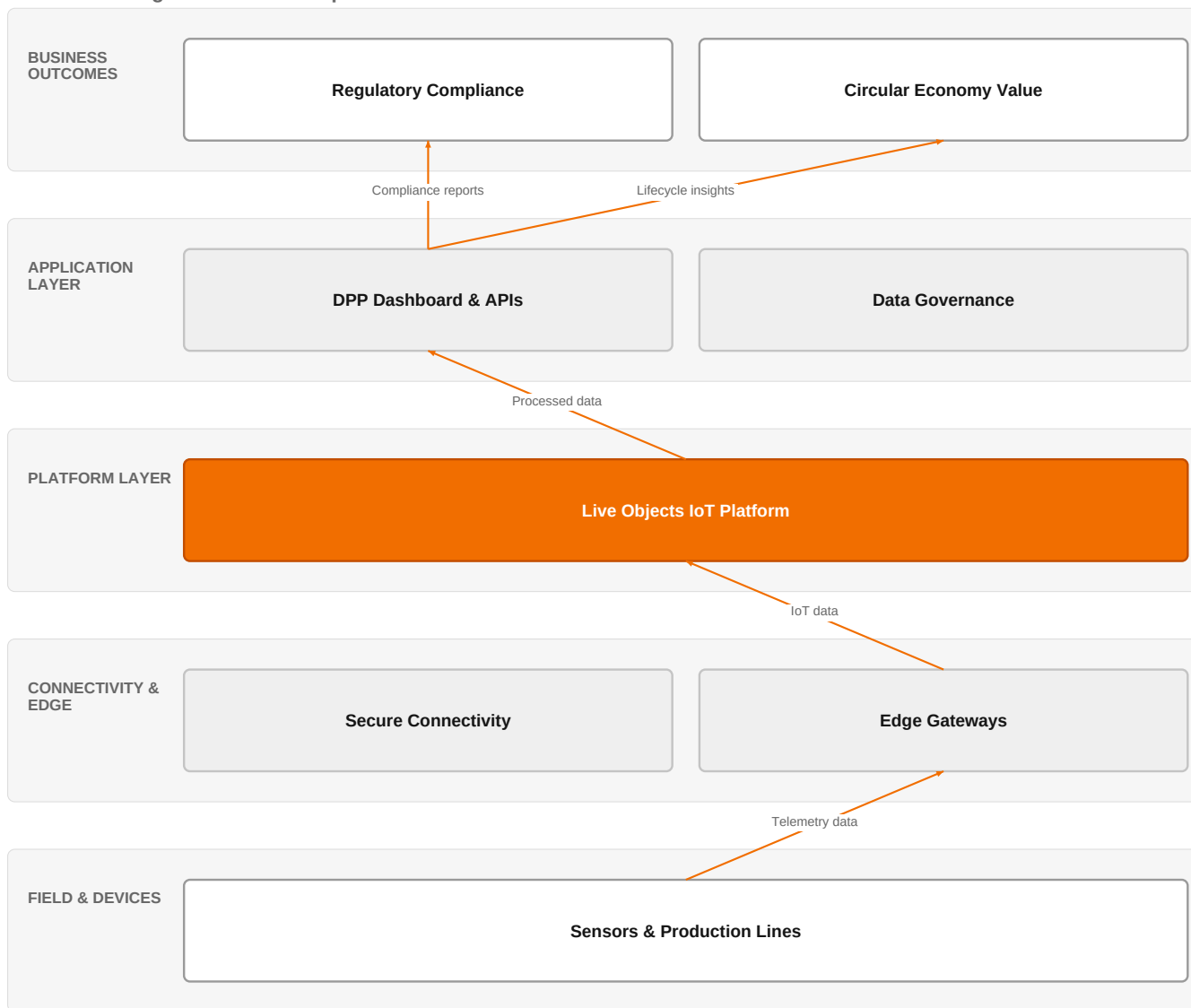
### Observed: tendered contract value, annualised

| Factor                                             | Value           | Basis      | Source                                | As of                  |
|----------------------------------------------------|-----------------|------------|---------------------------------------|------------------------|
| Tendered contract value in matching CPV categories | €20.9m per year | observed   | TED (Tenders Electronic Daily) · ted  | 2026-05-19..2026-08-05 |
| Assumed obtainable share of the serviceable market | 2.8 % of SAM    | assumption | config/sizing.yaml · obtainable_share | size-2026-08-a         |

**Read this before quoting the number.** Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

## The solution, in one picture

### IoT-Enabled Digital Product Passport Architecture



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This architecture shows how Orange's Live Objects platform securely collects and manages IoT data to generate digital product passports, enabling regulatory compliance and circular economy value.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

## What has changed

The EU's Ecodesign for Sustainable Products Regulation (ESPR, EU 2024/1781) mandates Digital Product Passports for electronics and appliances by 2027, creating an urgent need for machine-readable product lifecycle data. Digital Product Passports are rapidly emerging as core infrastructure for sustainable manufacturing, enabling lifecycle transparency, circular economy practices, and regulatory compliance. However, effective DPP implementation requires robust integration of technologies like IoT to overcome scalability and data integrity challenges.

Sources: SIG-BA196FF05244 Springer Science and B 2026-02-03; SIG-36FFB2435F8A openalex.org 2026-04-20; SIG-F47C31AB60CB MDPI AG 2026-02-14

## Who buys, and why now

The chief sustainability officer or head of compliance typically signs, but the head of manufacturing operations feels the pain of manually compiling lifecycle data across fragmented supply chains. The trigger is the approaching ESPR deadline, forcing them to act now to avoid non-compliance penalties and to meet customer demands for transparency. They need a way to automatically capture and share product data from the factory floor to the recycler, without disrupting production.

## Why it is hot now — every claim bound to a dated source

- Digital Product Passports enable lifecycle transparency, circular economy practices, and regulatory compliance in sustainable manufacturing. [SIG-36FFB2435F8A]
- The Digital Product Passport has emerged as the centralized data framework for the Circular Economy, requiring integration of technologies like IoT to overcome scalability and data integrity challenges. [SIG-F47C31AB60CB]

## What Orange would deliver, and why it can win

### What Orange would deliver

Orange would assemble a solution using its Live Objects IoT platform to collect and manage product lifecycle data from sensors and production systems. Orange IoT experts and Digital experts would design the data model and integration with existing ERP and MES systems. Orange Group researchers could contribute to interoperability standards. The engagement would start with a discovery workshop to map data sources and compliance requirements, then deploy Live Objects to ingest and manage data, and finally build dashboards and APIs for DPP generation and sharing with supply chain partners.

### Why Orange can win

Orange brings a unique combination of IoT platform expertise (Live Objects), deep security certifications (ISO 27001, TISAX), and a track record with industrial clients like Saint-Gobain Glass and Colgate-Palmolive. The ability to deliver on sovereign, certified infrastructure aligns with the trust and sovereignty axis that is central to Orange's strategy. However, the assets are thin on specific DPP domain knowledge, so Orange would need to partner for regulatory expertise.

### Named assets behind this — each individually inspectable (LK-08)

| Asset                        | Type            | Link | Why it is linked                                           | Curator     |
|------------------------------|-----------------|------|------------------------------------------------------------|-------------|
| Live Objects                 | offer           | L1   | offer provides the technology in a matching domain         | unconfirmed |
| ISO 27001                    | certification   | SUP  | certification Orange holds is required by this vertical    | unconfirmed |
| TISAX                        | certification   | SUP  | certification Orange holds is required by this vertical    | unconfirmed |
| IoT experts                  | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Digital experts              | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Defense and security experts | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Orange Group researchers     | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Colgate-Palmolive            | reference       | SUP  | published reference in this vertical, different use case   | unconfirmed |
| Mondelez International       | reference       | SUP  | published reference in this vertical, different use case   | unconfirmed |
| Saint-Gobain Glass           | reference       | SUP  | published reference in this vertical for this use case     | unconfirmed |
| Skytale                      | reference       | SUP  | published reference in this vertical, different use case   | unconfirmed |

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

## Competitive landscape

**HIGH** competitive intensity (score 8.2; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

| Competitor                   | Category              | Position here                                                                                         | Basis      | Evidence |
|------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------|------------|----------|
| AWS                          | Hyperscaler           | sells IoT platform; competes in OX: Smart Industries — also an Orange partner                         | structural | —        |
| Deutsche Telekom / T-Systems | Telecom operator peer | sells IoT platform; active in Manufacturing; competes in Connectivity Solutions, OX: Smart Industries | structural | —        |

|                   |                           |                                                                                                       |            |   |
|-------------------|---------------------------|-------------------------------------------------------------------------------------------------------|------------|---|
| NTT DATA          | Global systems integrator | sells IoT platform; competes in Connectivity Solutions, OX: Smart Industries                          | structural | — |
| Proximus          | Telecom operator peer     | sells IoT platform; competes in Connectivity Solutions                                                | structural | — |
| Telefonica Tech   | Telecom operator peer     | sells IoT platform; competes in Connectivity Solutions                                                | structural | — |
| Telia Company     | Telecom operator peer     | sells IoT platform; competes in Connectivity Solutions, OX: Smart Industries                          | structural | — |
| Verizon Business  | Telecom operator peer     | sells IoT platform; competes in Connectivity Solutions                                                | structural | — |
| Vodafone Business | Telecom operator peer     | sells IoT platform; active in Manufacturing; competes in Connectivity Solutions, OX: Smart Industries | structural | — |

**evidenced** means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

## Qualifying questions for the first meeting

- 1 What is your current process for collecting and sharing product lifecycle data across your supply chain?
- 2 Which product lines are in scope for the EU Digital Product Passport regulation, and what is your timeline for compliance?
- 3 Do you have existing IoT sensors or connectivity on your production lines or products?
- 4 Who in your organization owns the data that would feed a digital product passport?
- 5 Have you already evaluated any DPP solutions or pilots? What were the outcomes?
- 6 What are your key concerns about sharing lifecycle data with customers, regulators, or recyclers?

## Objections you will meet

| They will say                                                 | You can answer                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| We already have an IoT platform from AWS or another provider. | That's true, and we can work with that. But our Live Objects platform is designed for secure, scalable IoT data management, and we can integrate with your existing systems. We also bring deep security certifications and industry expertise that can accelerate your DPP compliance.           |
| The ESPR deadline is still a few years away; we can wait.     | You're right, but the regulation is already adopted, and the data infrastructure needed for DPPs takes time to build. Starting now allows you to pilot, refine, and scale without the pressure of a last-minute rush. Early movers also gain a competitive advantage in sustainability reporting. |
| We don't have the budget for a new platform.                  | We understand budget constraints. Our solution can be phased, starting with a pilot on one product line to demonstrate value and ROI. We can also leverage existing investments in IoT and data platforms to minimize additional costs.                                                           |

## Risks and unknowns

The main risk is that the ESPR deadline slips, reducing urgency. There is also uncertainty about the final DPP data standards and interoperability requirements, which could change the solution design. Additionally, manufacturers may be reluctant to share sensitive lifecycle data with partners, requiring robust data governance. Finally, the cost of retrofitting sensors to existing products may be prohibitive for some customers.

## Next action, by role (FR-17)

| Role                   | Do this next                                                                                                                                                                                                                                                                                                 |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Presales / Proposal    | Prepare a bid brief that positions Live Objects as the IoT backbone for a Digital Product Passport solution, leveraging ISO 27001 and TISAX certifications to address data integrity and security, and citing the Saint-Gobain Glass reference for credibility.                                              |
| Sales                  | In a meeting with a manufacturing client's operations or sustainability leader, say: 'We've helped global manufacturers like Colgate-Palmolive and Saint-Gobain Glass use IoT to improve operational visibility; we can apply that to make your product lifecycle data ready for Digital Product Passports.' |
| Strategist / Innovator | Commission a deep-dive brief on how Orange's Live Objects IoT platform plus Digital experts can address the scalability and data integrity challenges of Digital Product Passports, using the Colgate-Palmolive, Mondelez International, and Saint-Gobain Glass references as proof points.                  |

## Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

| Signal           | Date       | Publisher                    | T | Title and link                                                                                                     |
|------------------|------------|------------------------------|---|--------------------------------------------------------------------------------------------------------------------|
| SIG-63476A345BB2 | 2026-06-01 | cordis.europa.eu             | 1 | <a href="#">Fostering Resilient and Auto-Compliant Manufacturing Value Chains through Advanced Digital Prod...</a> |
| SIG-36FFB2435F8A | 2026-04-20 | openalex.org                 | 1 | <a href="#">A Comprehensive Review of Digital Product Passports in Sustainable Manufacturing: Current State...</a> |
| SIG-F47C31AB60CB | 2026-02-14 | MDPI AG                      | 1 | <a href="#">Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...</a> |
| SIG-BA196FF05244 | 2026-02-03 | Springer Science and Busi... | 1 | <a href="#">Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...</a> |
| SIG-7B174A2FD055 | 2026-01-29 | MDPI AG                      | 1 | <a href="#">Verisav Vocabularies: RDF/OWL Ontologies for Digital Product Passport and After-Sales Service A...</a> |
| SIG-4FE21FD2ED06 | 2026-01-01 | openalex.org                 | 1 | <a href="#">On the Utilization of Digital Product Passports for Intelligent Metal Sorting and Recycling</a>        |
| SIG-53ADAC7DA5A5 | 2026-01-01 | RSC Sustainability           | 1 | <a href="#">Supply chain optimisation using physics-informed machine learning for digital product passport</a>     |
| SIG-DA2C77C45DF6 | 2026-01-01 | Fashion Works                | 1 | <a href="#">The Digital Product Passport</a>                                                                       |
| SIG-8123897AE6E6 | 2025-12-15 | openalex.org                 | 1 | <a href="#">Bridging digital tools and human skills: A sustainable competence framework for implementing di...</a> |
| SIG-2B47C056B7E5 | 2025-10-30 | openalex.org                 | 1 | <a href="#">Digital material passports in the AEC industry: a scoping review and conceptualisation of futur...</a> |
| SIG-EFEBD520406D | 2025-10-15 | openalex.org                 | 1 | <a href="#">Stakeholder perspectives on digital product passports for construction products</a>                    |
| SIG-40DC35D6534F | 2025-10-14 | openalex.org                 | 1 | <a href="#">Predictive Generation of Digital Product Passport as Decision Support in Circular Networks</a>         |
| SIG-6FBECA0F7EA7 | 2025-09-10 | openalex.org                 | 1 | <a href="#">Design of a Sensor-Based Digital Product Passport for Low-Tech Manufacturing: Traceability and ...</a> |
| SIG-D4A38DBA082C | 2026-06-30 | Engineering News             | 2 | <a href="#">Prepare for the future of trade: Understanding Digital Product Passports - Engineering News</a>        |

## How this brief was made

**Computed, not written.** Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

**Written, then checked.** The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive\_landscape (no claim survived evidence binding); diagram (3 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

| What                        | Version                     | When                      |
|-----------------------------|-----------------------------|---------------------------|
| Scores — weight set         | w-2026-08-a                 | 2026-08-18T19:08:26+00:00 |
| Market sizing — assumptions | size-2026-08-a              | 2026-08-18T21:06:15+00:00 |
| Competitor register         | competitors-2026-08-a       | 2026-08-18T21:06:18+00:00 |
| Narrative — prompt / model  | describe-v1 / deepseek-chat | 2026-08-18T20:40:41+00:00 |
| Pipeline                    | 0.1.0                       | 2026-08-18                |

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.